TorqCapture — Zone-Capture Tech v1

*2026-09-06. The prototype **gate** for TorqCapture — the same role the Collapse Test Protocol plays for TorqSiege. Electronic scoring is LOCKED ([[TorqCapture — Design Study v1]] decision 3): the game is only as good as the reader that scores it. This doc states the one gating unknown, analyses the three real technologies with honest pros/cons + India-available modules + INR prices, **recommends a primary + a fallback**, specs the build (per-car tag → per-zone reader → controller → LED + screen) with a costed BOM, and gives a cheap **bench test** to prove it before a rupee goes to the full event. Don't build TorqCapture until the bench test passes. Prices from a Sept-2026 India-market scan (sources at the bottom).*

The question we're answering

Not "can a reader detect a tag" (yes, trivially). The real bar:

> **Can a zone reader detect + count how many cars of each team are inside a ~1.5–2 m ring — and ONLY that ring — reliably, live, while the cars shove, overlap and pile up?**

That is the crux, because the entire rulebook ([[TorqCapture — Rulebook]] §2) rests on a live, correct **count per team per zone**. Three things make it hard, and any candidate tech must beat all three:

1. **Confinement.** The reader must see cars **inside** the 1.5–2 m ring and not a car 1 m outside it, or in the next zone 4–6 m away. A reader that "sees" too far can't tell A from B.
2. **Multi-tag / multi-car counting.** 2–3 cars per team can be crammed into one ring, overlapping and touching. The reader must count *all* of them, per colour, not just the nearest or the strongest.
3. **Live + robust under chaos.** Cars are moving, shoving, flipping; tags are on moving bodies; the count updates several times a second and can't drop a car just because it got shoved to the ring edge for half a second.

NFC is ruled out — and why. NFC reads at **centimetres** (it's the tap-to-pay range). A car would have to sit almost on top of the reader to register, and it couldn't cover a 1.5–2 m ring or count multiple cars spread across it. NFC solves "is this one card touching this one pad", not "how many cars are in this circle". Off the table. The real candidates are **UHF RFID, BLE/RSSI beacons, or a computer-vision overhead cam.**

Option analysis (the three real candidates)

A. UHF RFID — a passive tag per car + a zone reader/antenna

Each car carries a cheap passive **UHF tag** (a sticker/inlay); each zone has a **UHF reader + antenna**. The reader interrogates the field several times a second and returns the list of tag IDs it sees — anti-collision protocols read many tags at once, which is exactly the multi-car count we need.

- **Pros:** tags are **near-free** (₹15–25 each) and passive (no battery on the car — ideal for a strap-on tag on a rented car); anti-collision multi-tag read is a solved, off-the-shelf feature (it's how toll and warehouse gates count pallets); per-tag ID means team colour = a tag range, and a 3-team variant is just three ID ranges.
- **Cons — the hard one is confinement.** UHF fields **spill.** A reader tuned for a 1.5–2 m ring will happily read a tag 3–4 m away or in the next zone unless you **turn the power down and shape the antenna** carefully — and near the floor, metal (car chassis, motors) and other cars **detune and shadow** tags, so a car can drop out mid-pile. Getting a field that fills the ring and stops at its edge is real RF tuning work, and two zones 4–6 m apart may cross-read each other. Read *rate* is fine; read *boundary* is the risk. Also: position within the ring is unknown (it's presence, not location), so you can't use it as the content feed.
- **India modules + INR (Sept 2026):**

Part	Module	~INR
Small integrated reader (TTL/USB, ceramic antenna, 0–6 m, **power-adjustable 15–26 dBm** — the tuning knob we need)	YRM100-class UHF module	~4,500–7,000 each (imported ~$48–56)
Fixed reader + external 9 dBi antenna (longer range — likely **too much** spill for a 2 m ring, listed for context)	India integrated UHF reader	~25,000–34,000 each
Passive UHF tag/inlay (per car)	Generic EPC Gen2 sticker	~15–25 each (falls to a few ₹ in bulk)

Three zones (Domination) = **3 readers** — the power-adjustable YRM100-class at ~₹4.5–7k each (~₹14–21k for 3), NOT the ₹25k+ fixed readers (their range is the wrong direction). Tags are a rounding error.

B. BLE / RSSI beacons — an ESP32 per car + an ESP32 scanner per zone

Each car carries a battery **BLE beacon** (an ESP32, or a coin-cell iBeacon); each zone has an **ESP32 scanner** that lists nearby beacons and estimates distance from **RSSI** (signal strength). "In the zone" = RSSI above a threshold.

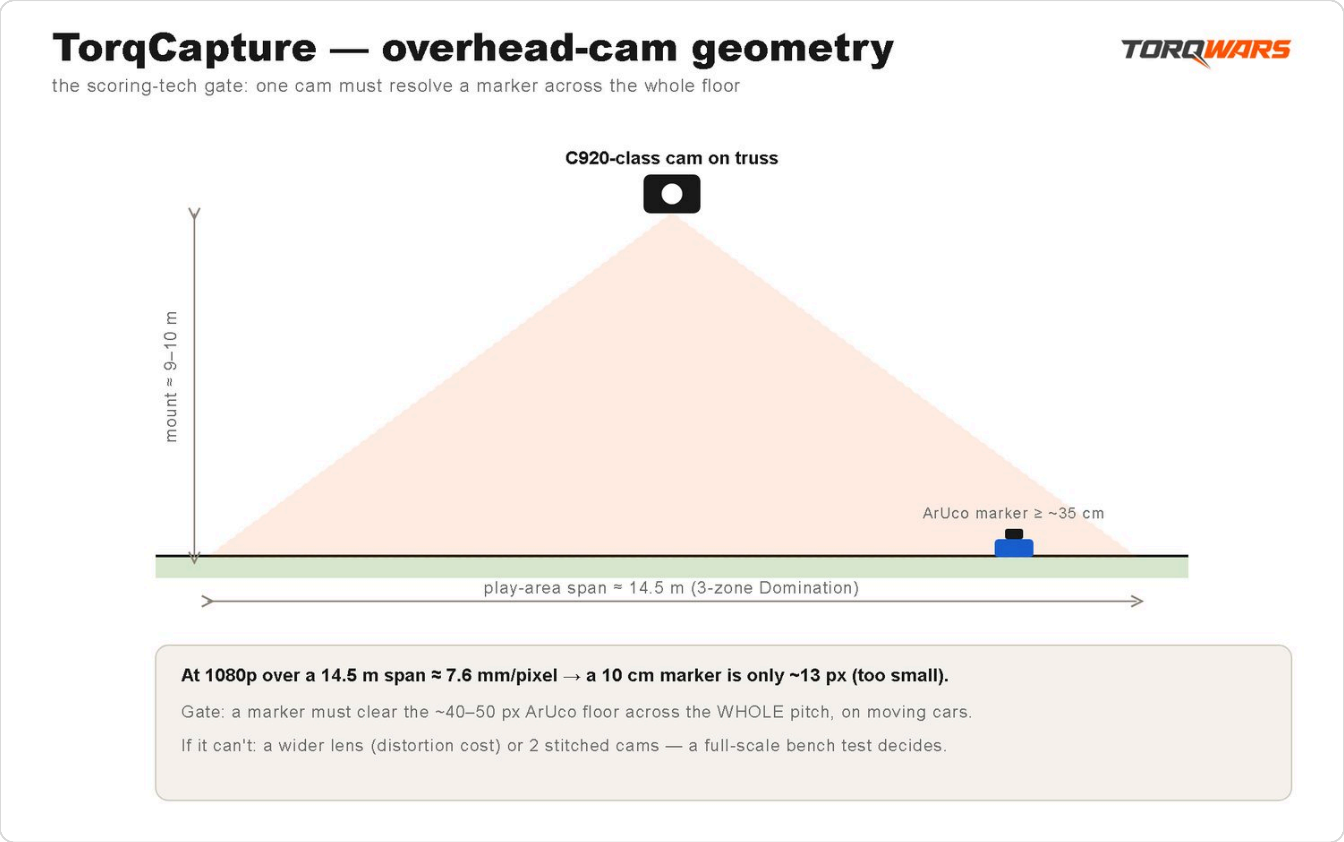
- **Pros:** ESP32 is dirt-cheap and we already use it across the kit; fully DIY and hackable; the same ESP32 can beacon *and* scan; battery beacon is a clean strap-on. Cheapest hardware by far.
- **Cons — RSSI is noisy.** Signal strength is a **terrible ruler.** It swings with orientation, a body between beacon and scanner, other cars, and floor reflections — the same car at the same spot can read "in" then "out" second to second. A hard 1.5–2 m boundary from RSSI is exactly where BLE is worst; you'd need heavy smoothing (which adds lag, bad for a shove-fest) and it will still flicker at the ring edge. Counting is fine (each beacon is a unique ID); **confinement is the weak point**, worse than UHF because you can't "shape" an RSSI field, only threshold it. Beacons need charging/coin-cells (a battery-management chore on rentals). No position, so not a content feed.
- **India modules + INR (Sept 2026):**

Part	Module	~INR
ESP32 beacon (per car)	ESP32 WROOM dev board	~250–600 each
ESP32 scanner (per zone)	ESP32 WROOM dev board	~250–600 each
(or) coin-cell iBeacon (per car)	Generic BLE beacon	~300–700 each

Cheapest option on paper (3 zones + 8 cars ≈ **₹3–7k of ESP32s**), but the one most likely to score *wrong* under a pile-up.

C. Computer-vision overhead cam — coloured / ArUco marker tops + one camera + software

One camera looks down at the floor. Each car wears a **coloured disc or an ArUco fiducial marker** on top (strap-on/velcro — nothing bolted). Software (OpenCV) finds each marker, reads its **exact (x, y) position**, and asks "is this marker inside ring A?" — pure geometry. The count per team per zone falls straight out of positions + colours.



- **Pros — it beats all three hard problems at once. Confinement is exact:** a ring is a circle in software; a car is in it or it isn't, to the pixel — no field to tune, no threshold to flicker, no cross-read between zones. **Counting is exact and per-team:** every marker is found and coloured, overlaps are resolved visually, 2–3 cars in one ring is trivial. **It doubles as the content feed** — the overhead cam *is* the money shot ([[TorqCapture — Rulebook]] §8: zones flipping colour on camera), so one device scores the game AND shoots the reels + drives the big-screen map. ArUco markers give a rock-solid unique ID per car; coloured discs are even simpler if we only need team + count. Software cost is zero (OpenCV, free) and it runs on the show laptop we already have.
- **Cons:** needs the camera **mounted high enough to see the whole play area** (a truss/pole overhead — we already rig one for content) with even-ish lighting; a marker fully **occluded** (car flipped upside-down, or another car parked on top) drops out until it's visible again — mitigated by markers on all-round-visible tops and a short "last seen" hold; it's a bit more software to get right than plugging in a reader. Calibration (mapping camera pixels to floor rings) is a one-time setup per event. One camera covers one floor — fine for a single-zone mode (Hill/Hotzone: one ring, camera low, marker close), **but the 3-ring Domination span is the real coverage risk and must be bench-proven before the camera is bought** (see the full-span trial in *The bench test*).

- **The Domination coverage math (why single-zone ≠ 3-zone for CV).** Domination's A–C axis is **~14.5 m wide** ([[TorqCapture — What The Event Needs]] §4 fit math: tight end, 1.5 m rings, ~4 m centers). A C920-class webcam has a **~70–78° horizontal FOV**, so to span 14.5 m it must sit **~9–10 m up** (width ≈ 2·h·tan(FOV/2) ⇒ h ≈ 14.5 / (2·tan(39°)) ≈ 9 m) — a truss height most 3000 sq ft halls **do not have.** Worse, at that height a **~10 cm ArUco marker is only reliably detected to ~3 m at 1080p** (web-checked rule of thumb — detection range scales with marker-pixels, ~marker_size × focal_px / min_readable_px), while the far ring sits **~9 m straight down and ~11 m slant** from an overhead cam — **~3–4× beyond the reliable read range. Cross-check via pixel density:** spanning the full 14.5 m axis in one 1920-px-wide 1080p frame gives **≈7.6 mm/pixel** (14,500 mm ÷ 1920 px) at the floor plane — a ~10 cm marker is only **~13 px** across at that density, well under the **~40–50 px per side** an ArUco marker typically needs for reliable pose/ID detection. Working the floor: a marker needs to be **~35–40 cm across** (≈40–50 px ÷ ~1.1–1.4 px/mm) just to clear that per-pixel floor at the far ring — on top of still needing the ~9–10 m mount height for FOV coverage in the first place. So a single 1080p cam that covers one small ring up close will very plausibly **fail to resolve markers across the full 3-ring span** — both the range-based estimate and the pixel-density cross-check land in the same place. The realistic mitigations, in cost order: (a) **bigger markers** (~25–35 cm discs, close to the ~35–40 cm the pixel-density calc wants — but they overhang a ~15–20 cm 1/16 buggy top, awkward on a rented car); (b) a **higher-res camera** (4K roughly doubles read range to ~6 m and halves mm/pixel to ~3.8 — still short of what 9–11 m needs); (c) **two stitched cameras**, each over ~half the span at a lower, marker-resolving height (a materially different OpenCV build — stitching/multi-cam merge code — costed as the contingency in [[TorqCapture — What The Event Needs]] §2); (d) run Domination as **fewer/bigger rings** (e.g. 2 zones, wider rings, shorter axis) or **marshal-scored**. This is exactly what the full-span bench trial below exists to decide **before** the camera buy — not at the T-3 rehearsal, after the money is committed.
- **India modules + INR (Sept 2026):**

Part	Module	~INR
Overhead camera (1080p, fixed FOV, USB)	Logitech C920-class webcam	~5,150–8,500 (one covers a single-zone footprint at a modest mount height; the full 3-ring Domination span needs a taller mount — or a second camera — see the coverage calc below)
Marker tops (per car)	Printed ArUco on card/foam + velcro, or a coloured disc	~20–50 each (we print)
Compute	The show laptop (already in the kit)	0
Software	OpenCV + a ~200-line script (we write)	0

Cheapest *scoring* option after the camera, because the camera is a content buy we'd make anyway and the compute + software are free.

Recommendation

Primary: Computer-vision overhead cam (Option C). Fallback: UHF RFID (Option A).

Why CV is the primary. It's the only candidate that solves **confinement** cleanly — a ring is a circle in software, so "in the zone / not in the zone" is exact geometry with no field to tune and no RSSI to smooth, and two zones 4–6 m apart never cross-read. It counts **per team, under a pile-up**, because it *sees* the cars. And it **doubles as the content feed** — the overhead flip-colour map is already the game's hero shot, so one device scores the match, feeds the big screen, and shoots the reels. That triple duty is decisive for a content-first brand: we were buying the overhead cam anyway. Bias toward "what actually confines to a zone and counts multiple cars" points straight at vision.

Why UHF is the fallback, not BLE. If CV struggles (bad venue lighting, no clean overhead mount, too much occlusion in dense piles), **UHF RFID** is the backup: passive battery-free tags are perfect for rented cars, and anti-collision multi-tag counting is off-the-shelf. Its weakness is confinement (field spill), but that's a *tuning* problem (drop the power on a YRM100-class reader, shape the antenna, space the zones) with a known solution — unlike BLE's RSSI noise, which is a *physics* problem you can only paper over. **BLE/RSSI is ruled out as a primary** for the same reason NFC is ruled out at the other extreme: it can't draw a reliable 1.5–2 m boundary. Keep BLE only as a cheap experiment if both CV and UHF somehow fail.

The hedge: the bench test (below) runs **CV and UHF side by side on the same taped ring** so the decision is made on real numbers, cheaply, before the event build. CV is the expected winner; UHF is the proven backup; the loser is dropped.

The build (once the bench test picks the tech)

The scoring pipeline is the same whichever reader wins — only the "reader" box changes.

Pipeline: `per-car tag/marker → per-zone reader (or 1 overhead cam) → controller → count → majority → hold-timer → capture → LED colour + score to the big screen.`

- **Per-car tag/beacon (strap-mounted — rented fleet, nothing bolted):**
- *CV:* a printed **ArUco marker or coloured disc** on a light foam/card top, velcro/zip-tied so it faces up and is visible all round. ~₹20–50/car, we print. Team colour = marker colour/ID range.
- *UHF:* a **passive UHF tag** stuck to a strap-on flap on top of the car (kept off the metal chassis with a foam spacer so it isn't detuned). ~₹15–25/car.
- Nothing bolted, screwed or glued — velcro/zip-tie, comes off clean at handback (same rule as the foam bump-rings, [[Car Fleet, Scale & Protection]]).
- **Per-zone reader:**
- *CV:* **one overhead camera** covers all zones at once (no per-zone hardware) for a single-ring mode; spanning the full 3-ring Domination footprint needs the mount height derived below (or a second stitched camera) — still the big cost saving over per-zone hardware once that's sized correctly.
- *UHF:* **one power-tuned reader + antenna per zone** (3 for Domination), each dialled to fill its ring and no further.
- **The controller (the brain):** an **ESP32 or the show laptop** runs the game logic for every zone: read the count per team → decide the majority (or contested) → run each zone's **hold-timer** → on completion **flip** the zone (ownership + colour) → **tick points** → push **LED colour** (WS2812B ring, §arena option 3) + the **live score to the big screen.** For CV this is the same laptop running OpenCV; for UHF it's an ESP32 aggregating the readers over serial/Wi-Fi. Mode presets (zone map, hold-timer, relocation interval, points cap) load per mode ([[TorqCapture — Rulebook]] §4).

- **Output:** WS2812B LED ring per zone (colour = owner, pulse = capturing) + a scoreboard on the projector/screen the MC calls off.

Parts BOM (feeds the costs stream — CV primary build). Split in two: the SCORING hardware (what actually reads the zones) and the LED rings, which are the optional content-hero (arena option 3, [[TorqCapture — Rulebook]] §5) and are costed separately because (a) they're a content upgrade, not required to score — a flat-taped or overhead-cam-defined ring (option 1) scores for ~₹0 — and (b) their length depends on how many rings the mode lights.

Scoring hardware (CV primary):

Part	Qty — why	~INR	Notes
Overhead camera (1080p USB)	1 — covers the play area at the mount height the coverage calc below requires (single-zone modes are comfortable; the 3-ring Domination span is the conditional case — see Option C cons and the full-span bench trial); also the content feed	5,150–8,500	Logitech C920-class; a buy we'd make for content anyway
Show laptop (compute)	1 — runs OpenCV + game logic + scoreboard	0	Already in the shared kit
OpenCV script	1 — marker detect → ring geometry → count → capture logic	0	We write (~200 lines)
ArUco/coloured marker tops	**one per rented fleet car — 12–16 + ~4 spares** (fleet-reconcile note below)	400–1,000 (lot)	We print on card/foam + velcro; ~₹20–50 each so qty barely moves cost
Cabling, USB extension, mounts, power	lot	1,000–2,000	Camera on the content truss; scoring runs on the free show laptop as controller — no paid controller in this subtotal
CV scoring subtotal (no LED rings)		**≈ 6,550–11,500**	One camera scores 1–3+ zones; scales cheap. **Excludes the LED-driver ESP32** — its only job is pushing colour to WS2812B strips, so in the flat-taped/no-LED config it isn't needed to score; it now sits in the LED block below and re-enters the cost only with the LED upgrade

LED rings — optional content-hero (arena option 3), priced from ring geometry:

A ring is spec'd at **~1.5–2 m across — that's a DIAMETER**, so each ring's circumference (the strip length that closes it) is **π × 1.5 to π × 2 ≈ 4.7–6.3 m of WS2812B per ring** — i.e. **more than one 5 m strip** per ring at the 2 m end. Cost therefore scales with how many rings the mode lights:

LED line	Qty — why	~INR	Notes
WS2812B strips — single-zone modes (Hotzone / The Hill / Siege Point / Courier)	1 ring ≈ 4.7–6.3 m = **1–2 × 5 m strips**	2,400–5,200	One ring, reused across every single-zone mode (add the ESP32 + PSU rows below once, not per mode)
WS2812B strips — Domination (3 rings A/B/C)	3 rings × 4.7–6.3 m ≈ **14–19 m = 3–4 × 5 m strips**	7,500–10,400	A single 5 m strip barely closes ONE 1.5 m ring and can't close a 2 m ring — three rings need **3–4 strips, not one**
ESP32 LED driver (**moved here from scoring** — only needed when LEDs are built)	1–2	250–1,200	Takes the per-zone owner/state from the laptop and pushes WS2812B colour; not required for flat-taped scoring, so it's costed with the LEDs, not the reader
5V PSU + power injection (for the 14–19 m Domination run)	1 set	1,500–4,000	60 LED/m addressable WS2812B pulls **tens of amps at brightness** across 14–19 m — needs a dedicated 5V supply (5V 30–40 A ≈ ₹1,500–3,000) + injection taps every few metres. The scoring cabling row does NOT cover this (it's scoped to the camera/USB/mount). Run the rings low-brightness / single-colour and current drops so a smaller existing supply can serve — that's the way to zero this line
LED-rings block (Domination) subtotal	strips + LED-driver ESP32 + 5V PSU	**≈ 9,250–15,600**	7,500–10,400 strips + 250–1,200 ESP32 + 1,500–4,000 PSU. The ESP32 + PSU are the LED block's **control + power**
CV build WITH Domination LED rings	scoring subtotal (6,550–11,500) + LED block (9,250–15,600)	**≈ 15,800–27,100**	Sum of the scoring subtotal and the LED-block subtotal; the LED is the content upgrade — drop to flat-taped rings (option 1) and the build is just the ₹6,550–11,500 scoring subtotal

UHF fallback (if CV fails — swaps the camera for per-zone readers; LED rings add on top, same as CV):

Part	Qty — why	~INR	Notes
Power-adjustable UHF reader + antenna	3 — one per zone	13,500–21,000	YRM100-class, ₹4.5–7k each
Passive UHF tags	**one per rented fleet car — 16–20 (12–16 cars + ~4 spares)**	300–500 (lot)	Near-free at ₹15–25 each

Part	Qty — why	~INR	Notes
UHF scoring subtotal (readers + tags only)		≈ 14,000–22,000	Readers + tags only — does NOT yet include the ESP32 reader-aggregator controller (UHF needs one to fuse the per-zone readers over serial/Wi-Fi; ₹250–1,200) + cabling/mounts (₹1,000–2,000) ≈ ₹1,250–3,200, nor the LED rings. The CV build needs no paid controller (its scoring runs on the free show laptop) and its subtotal already counts its own cabling — so add controller + cabling to UHF for a like-for-like number ≈ ₹15,250–25,200

Fleet-reconcile note (why 12–16 markers, not 8): only **6–8 cars are live per heat**, but the rented fleet is **12–16 cars** (6–8 live + charging rotation + spares — see [[Car Fleet, Scale & Protection]] §1 and [[TorqFall — What The Event Needs]] §1, which the Rulebook §3 explicitly reuses). Because a charged car is **swapped in at each ~30–60 s reset** (Rulebook §6.8, §10) and the marker/tag is strap-mounted (velcro/zip-tie, nothing bolted — [[Car Fleet, Scale & Protection]]), **every car in the rotation carries its own marker/tag** so it's ready the instant it rolls on — we do NOT unstrap a marker off a spent car and re-strap it onto a charged one mid-turnaround (that would add a fiddly step to a 30–60 s reset and risk a dead-tag car on the floor). One marker per fleet car = **12–16 + ~4 spares**; at ₹20–50 (CV) or ₹15–25 (UHF) each the qty bump is a rounding error.

CV is both the cheaper scoring build *and* the content feed — the reason it's primary on cost as well as accuracy (its scoring subtotal ≈₹6.55–11.5k undercuts UHF's ≈₹14–22k, and the LED rings are a tech-agnostic add-on that lands on either build equally). Note the comparison is conservative in UHF's favour: the CV subtotal runs its controller on the **free show laptop** (₹0) and counts its own cabling/mounts (₹1,000–2,000), while the UHF ≈₹14–22k is readers + tags only and still needs a **paid ESP32 reader-aggregator** (₹250–1,200) plus its own cabling (₹1,000–2,000) — add that ≥₹1,250–3,200 to make it like-for-like and UHF rises to ≈₹15.25–25.2k, which only widens CV's lead. Full event costing (fleet, turf, show, crew) lives in [[TorqCapture — What The Event Needs]]; this BOM is the zone-tech line that feeds it.

The bench test (the cheap gate — do this before the event build)

Same spirit as the collapse test: a kitchen-table afternoon, near-zero cost, that returns a yes/no on the gating question before a rupee goes to the full build.

What you need (have-now / cheap):

- **2–4 RC cars** you can get today (borrowed/owned — don't wait for the rented fleet), each with a strap-on marker: for CV a printed ArUco/coloured disc on top; for UHF a passive tag on a foam-spaced flap. Split them into two "teams" by colour/ID.
- **One taped ring** on the floor, **1.5–2 m** across (gaffer tape).
- **CV rig:** a phone or a webcam mounted overhead (chair/ladder/pole) + a laptop running the OpenCV marker-detect script.
- **UHF rig (to compare):** one power-adjustable YRM100-class reader + laptop, tags on the same cars.
- A tape measure, a marker, a notebook (or a logging script).
- **For the full-span trial (the Domination coverage gate):** a way to mock the ~14.5 m 3-ring Domination footprint (three taped rings on the real A–C spacing) and to hoist the **one** camera to a realistic venue truss height (the tallest ladder/pole/mezzanine you can get — if the real hall's truss is

known, use that exact height). This is the single most important CV trial and the reason it's run **before** the camera is bought — the camera bought for content is the same one that must span three rings.

Method (run CV and UHF on the same ring, back to back):

1. Place N cars in/around the ring in a known layout; read the reported count per team off the screen; note right/wrong.
2. **Confinement test:** put a car **just outside** the ring (10–30 cm out) — does the reader correctly say "out"? Move it to the far edge inside — "in"? Walk a car from 1 m away straight through — does it flip in/out at the tape line, cleanly, once?
3. **Count test:** cram **2–3 cars per team** into the ring, overlapping/touching — does it count all of them, per colour, correct?
4. **Chaos test:** drive the cars, shove them, flip one — does the count track live without dropping a car for more than a moment? Time the lag.
5. **Cross-zone test (if two rings available):** two rings 4–6 m apart — does a car in ring A ever get counted in ring B?
6. **FULL-SPAN COVERAGE TEST (the Domination gate — the one CV trial that must pass before the camera is bought).** Tape **three** rings on the real Domination A–C spacing (~14.5 m axis at the tight end, [[TorqCapture — What The Event Needs]] §4), hoist the **single** camera to a realistic venue truss height, and confirm the marker in **ring A AND ring C** (the far rings, ~11 m slant from an overhead cam) is **resolved and correctly placed inside its ring**, not just the near/centre ring B. If the far-ring markers don't read at the height a real hall can give: try the mitigations in order — (a) bigger markers, (b) a higher-res (4K) camera, (c) a **second stitched camera** over the other half of the span, (d) accept Domination runs as **fewer/bigger rings or marshal-scored** and note it. Record which mitigation, if any, makes A and C both read. **This is the risk the T-3 rehearsal would otherwise surface too late — after the camera is bought at T-13; running it here de-risks the spend.**
7. Log every trial: tech, layout, reported vs true count, confinement pass/fail, lag, any flicker, **and the full-span far-ring read result + the height it needed**. ~20–30 trials covers it in an afternoon.

What to tune (the knobs):

- **Ring size** (1.5 vs 2 m — bigger reads easier, smaller is a tighter fight).
- **Reader range / confinement** (CV: camera height + ring geometry; UHF: reader power + antenna shaping + zone spacing).
- **Hold-timer** (~5 s default — long enough to be contestable, short enough to flow).
- **Marker/tag placement + a raised foam ring** — does a raised lip help cars hold position (rulebook §5 option 2) without occluding a top marker or shielding a tag?
- Smoothing / "last-seen" hold so a momentarily-occluded or shoved-to-edge car doesn't flicker out.

Pass / fail (the decision):

- **PASS** = one tech (expected: CV) gives the **correct per-team count** with **clean confinement at the ring edge** and **counts 2–3 overlapping cars per team**, **live under shoving with sub-second lag**, **no cross-zone bleed**, AND — for CV as the Domination scorer — **the full-span coverage test reads BOTH far rings (A and C) at a truss height the real hall can give** (trial 6). → lock that tech, freeze the ring size + hold-timer + the proven camera height/marker size (and whether Domination needs one cam or two), build the pipeline, grade TorqCapture in [[Game Ideas - Graded]], cost it in [[TorqCapture — What The Event Needs]], build the event.

- **CONDITIONAL** = it works but needs a constraint (e.g. CV needs a specific camera height / lighting / marker size, **or CV spans the full 3-ring Domination footprint only with a second stitched camera** (carry that BOM contingency), **or Domination runs on fewer/bigger rings or marshal-scored while single-zone modes stay CV**, or UHF needs zones ≥6 m apart and a raised non-metal tag mount — a spacing that overruns the owned turf roll, so it also forces an added-turf buy or a reduced-ring Domination, [[TorqCapture — What The Event Needs]] §4). → buildable, but bake the constraint into the arena spec + the run plan.
- **FAIL** = neither CV nor UHF holds a reliable boundary + count under a pile-up. → don't build electronic scoring as-is; fall back to a **marshal-scored** version (bigger/fewer zones, a human calling majority — slower, less content, but the game still runs) while the tech is reworked, and revisit before committing the full build.

Do-it checklist: (1) grab 2–4 cars + print some markers + tape one 1.5–2 m ring today; (2) run the CV rig (phone overhead + OpenCV) and, on the same ring, the UHF rig; (3) run the confinement + count + chaos + cross-zone trials, log them; (4) if PASS, lock the tech + ring + timer into [[TorqCapture — Rulebook]] §2–§5 and the BOM above into [[TorqCapture — What The Event Needs]]; (5) report the winning tech + the tuned numbers back here. Pairs with [[TorqCapture — Rulebook]] · [[Car Fleet, Scale & Protection]] (the strap-on rule) · [[Event Ops Standards (all games)]].

Price sources (Sept 2026): *ESP32 dev boards ₹250–600 (Robokits ₹284, Robu SmartElex; Zbotic 2026 guide ₹300–600); YRM100-class integrated UHF module ~$48–56 imported ≈ ₹4,500–7,000 (Alibaba/AliExpress/eBay listings), India fixed integrated readers ₹25,298–34,490 (Identium/IndiaRFIDStore) — noted as too-long-range for a 2 m ring; passive UHF tags/inlays ₹15–25/pc small-qty, falling in bulk (IndiaMART; global $0.05–0.25/unit, CPCON/JN 2026); Logitech C920-class 1080p webcam ₹5,150–8,500 (IndiaMART/Computech/Moglix/LT Online); WS2812B 5 m addressable strip ₹2,400–2,600 (Flipkart mdblck ₹2,499, Robu, Amazon pack ₹2,598). OpenCV/ArUco = free, open-source.*

Verified + changelog (2026-09-06): *first build. Stated the gating question (count + confine to a 1.5–2 m ring under shove/overlap) and ruled NFC out (cm range) with the reason. Analysed UHF RFID / BLE-RSSI / CV with honest pros-cons + India modules + INR. Recommended* ***CV primary, UHF fallback, BLE ruled out as primary*** *— reasoning: CV alone solves confinement (geometry, not a tuned/thresholded field), counts per-team under a pile-up, and doubles as the content feed; UHF's confinement is a solvable tuning problem, BLE's RSSI boundary is unsolvable physics. Specced the strap-on tag → per-zone reader (or 1 overhead cam) → controller → LED+screen pipeline and a costed BOM: CV* ***scoring*** *subtotal ≈ ₹6.8–12.7k (incl. the content camera), UHF scoring fallback ≈ ₹14–22k, LED rings split out as the optional content-hero (arena option 3).* ***Correction (2026-09-06, exam pass):*** *the earlier LED line was geometrically wrong — it treated one 5 m strip as enough for all three Domination rings. A ring is ~1.5–2 m diameter, so its circumference is π×1.5 to π×2 ≈ 4.7–6.3 m of strip PER ring; three rings need ~14–19 m =* ***3–4 × 5 m strips ≈ ₹7,500–10,400****, not ₹2,400–2,600. LED rings are now priced from ring geometry (1 strip-set for single-zone modes, 3–4 for Domination) and separated from the base; CV-with-Domination-LED total set to the sum of its component rows ≈ ₹14,300–23,100 (scoring 6,800–12,700 + 3-ring LED 7,500–10,400).* ***Also corrected:*** *marker/tag qty raised from 8 to* ***one per rented fleet car (12–16 + ~4 spares)*** *to match the 12–16-car fleet the Rulebook §3 reuses, with a fleet-reconcile note stating each rotation car keeps its own strap-on marker (no mid-reset marker-swap). Wrote the cheap bench test (2–4 cars, one taped ring, CV vs UHF side by side) with confinement/count/chaos/cross-zone trials, the tuning knobs (ring size, range, hold-timer, raised ring)* *and PASS/CONDITIONAL/FAIL bars. All hold-timer/ring/relocation numbers flagged as bench-test tunables, matching the rulebook.* ***Correction (2026-09-06, examiner pass 2):*** *(1) the CV-with-Domination-LED total read ≈₹14–22k but its two component rows sum to 14,300–23,100 — the ceiling understated the real high end by ~₹1,100; total corrected to ≈₹14,300–23,100 and shown as an explicit sum of the two rows. (2) The CV-vs-UHF cost line compared non-like-for-like subtotals: the CV subtotal includes the shared controller (₹250–1,200) + cabling/mounts (₹1,000–2,000) while the UHF ≈₹14–22k is readers + tags only — a founder totalling costs could under-count UHF. Added a note (in the UHF row and the closing sentence) that adding the same controller + cabling (≥₹1,250–3,200) to UHF makes it ≈₹15.25–25.2k like-for-like, which only widens CV's advantage; the "CV undercuts UHF" conclusion is unchanged.*

Correction (2026-09-06, costs-stream pass): *(1)* ***Mis-bucketed part fixed*** *— the LED-driver ESP32 (₹250–1,200) was inside the CV* ***scoring*** *subtotal, but its only job is pushing colour to WS2812B strips, so it isn't needed for flat-taped/no-LED scoring. Moved it into the LED-rings block; CV scoring subtotal drops* ***₹6,800–12,700 → ₹6,550–11,500****, and it re-enters the cost only with the LED upgrade. (2)* ***LED power added*** *— the LED block priced only the addressable strips with no 5V supply; 14–19 m of 60 LED/m WS2812B pulls tens of amps at brightness, so added a* ***5V PSU + power-injection*** *line (₹1,500–4,000; zero-able by running rings low-brightness/single-colour on a smaller existing supply). LED-rings (Domination) block subtotal is now strips + ESP32 + PSU ≈* ***₹9,250–15,600****, and CV-build-WITH-Domination-LED = scoring (6,550–11,500) + LED block (9,250–15,600) ≈* ***₹15,800–27,100*** *(was ₹14,300–23,100). (3)* ***Like-for-like wording corrected*** *— with the ESP32 out of scoring, the CV build runs its controller on the free show laptop (₹0) and counts its own cabling; UHF still needs a paid ESP32 reader-aggregator + cabling (₹1,250–3,200) → like-for-like ≈₹15.25–25.2k, CV's lead unchanged. Mirrored in [[TorqCapture — What The Event Needs]].*

Correction (2026-09-06, run-plan stream pass — CV full-span coverage gate): *the bench test proved confinement/count on* ***one*** *1.5–2 m ring but never tested whether* ***one*** *1080p overhead cam can resolve small top-markers AND confine across the* ***full 3-ring Domination span (~14.5 m)*** *— CV's single biggest feasibility risk, and the one that only surfaced at the post-spend T-3 rehearsal. Added: (a) the* ***Domination coverage math*** *to Option C cons — a ~70–78° FOV cam needs ~9–10 m truss to span 14.5 m (higher than most halls give), and a ~10 cm ArUco reads reliably only to ~3 m at 1080p while the far ring sits ~9 m down / ~11 m slant, so a single 1080p cam very plausibly fails across the span; mitigations in cost order = bigger markers → 4K cam → second stitched camera (a different OpenCV build) → fewer/bigger rings or marshal-scored Domination; (b) a* ***FULL-SPAN COVERAGE TEST (trial 6)*** *to the bench method — mock the three rings on the real ~14.5 m axis, hoist the one camera to a realistic truss height, confirm BOTH far rings A and C read,* ***before the T-13 camera buy****; (c) the full-span far-ring read as a* ***PASS gate*** *and the second-stitched-cam / reduced-ring / marshal-scored branches as* ***CONDITIONAL*** *outcomes. Mirrored as a costed second-camera contingency in [[TorqCapture — What The Event Needs]] §2 and as a pre-buy gate in [[TorqCapture — Full Run Plan]] THE GATE. Also noted in CONDITIONAL that a UHF ≥6 m-spacing fallback overruns the owned turf roll (added-turf buy or reduced-ring Domination). No CV/UHF/LED price rows changed by this pass (the second camera is a named contingency, not a base-build line).*

Correction (2026-09-06, examiner pass — residual over-claim): *three leftover lines still said flatly "one camera covers the whole floor" (Option C's India-modules table, the per-zone-reader pipeline description, and the scoring BOM row), contradicting the coverage-risk math already worked out above them. Reworded all three to state coverage is conditional on mount height (comfortable for single-zone modes, the 3-ring Domination span being the real risk case) and to point at the coverage calc / Option C cons / full-span bench trial rather than asserting blanket coverage. Also extended the coverage calc* *itself with a **pixel-density cross-check**: spanning the 14.5 m Domination axis in one 1920-px 1080p frame gives **≈ 7.6 mm/px**, so a 10 cm marker is only **~13 px** across — well under the ~40–50 px/side ArUco typically needs — meaning a marker would need to be **~35–40 cm** across to clear that floor, independently landing on the same bigger-marker range mitigation (a) already listed. No PASS/CONDITIONAL/FAIL bar or price changed — the full-span coverage trial (bench-test trial 6) was already a named PASS gate from the prior pass; this pass only removes the contradictory blanket-coverage wording and adds the mm/pixel + marker-size numbers behind it.*